

Program : Diploma in Computer Application and Business Management	
Course Code : 4253	Course Title: Relational Database Management System
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To provide a thorough knowledge in Relational Database management system.
- To develop programming skills in database, SQL, Relational algebra, triggers etc. which will help them to solve real world problems.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge on Database- MS Access		Office Automation Software	I

Course Outcomes:

On completion of the course, students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain database, system architecture, data models	16	Understanding
CO2	Apply relational model concepts, relational algebra, database design	15	Applying
CO3	Build SQL Operations	13	Applying
CO4	Illustrate Advanced SQL statements & explain transaction management	14	Understanding
	Series Exam	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain database, system architecture, data models		
M1.01	Understand Database	4	Understanding
M1.02	Know Database system structure(Architecture)	6	Understanding
M1.03	Understand Data models	6	Understanding
Contents: Define Field, Record, Database, Distinguish between Physical record and Logical Record, Explain Advantages of DBMS, Understand Schema, Subschema - Instances, Three schema architecture, Define data independence. Explain Storage manager, Query processor, Disk storage, Database Users, Explain various Database system application architectures like Centralized DBMS architecture, Basic client/server architecture, Two tier client/server architectures, and three tier client/server architecture, Understand Data models, Explain ER model, Define Entity, Attribute, Keys, Relationship types, Relationship set, Define Primary key, Candidate key, Super key, Explain ER diagram, Weak entity set, Describe EER Model: Subclasses - Super classes - Inheritance - Specialization - Generalization.			
CO2	Apply relational model concepts, relational algebra, and database design.		
M2.01	Understand Relational model Concepts	2	Understanding
M2.02	Explain Relational algebra operations	3	Understanding
M2.03	Perform Relational database design	5	Applying
M2.04	Explain functional dependency and normal forms	5	Applying
	Series Test-I	1	

Contents: Relational model - Concepts - Constraints - Relational algebra operations - Unary & Binary relational operations, Relational database design - Functional dependency - Normalization - 1NF - 2NF - 3NF - Boyce Codd Normal Form - 4NF - Join dependencies - 5NF.			
CO3	Build SQL Operations.		
M3.01	Understand SQL	5	Understanding
M3.02	Implement SQL using different commands	8	Applying
Contents: Explain Data types in SQL, Differentiate Domains, attributes, and tuples, Creation of Table, Explain CREATE TABLE command, Explain DROP TABLE Command, ALTER TABLE, Explain commands for creating and roping indexes, Explain SELECT statements with WHERE, ORDER BY clause with examples - Describe UPDATE Statement, INSERT, DELETE with example - Explain Joining tables - aggregate functions, - Explain BEGIN TRANS, COMMIT, ROLL BACK, SAVE POINT - Explain Views: Advantages, Creation and Updating- Explain Query optimization.			
CO4	Illustrate Advanced SQL statements & explain transaction management.		
M4.01	Understand advanced SQL	4	Understanding
M4.02	Understand Embedded SQL	4	Understanding
M4.03	Understand Transaction management	6	Understanding
	Series Test-II	1	
Contents: Understand advanced SQL - Define triggers and cursors - Explain reports and forms - Explain stored procedures and functions - Understand about Embedded SQL statements and its application - Explain dynamic SQL by specifying queries at run time - Understand Transaction management. Define Transaction - Explain transaction concepts - Differentiate Properties of transactions, Transaction Sates - Explain Concurrent Executions - Explain Serializability			

Text / Reference:

T/R	Book Title/Author
R1	Elmasri, Navathe, Somarajulu, and Gupta, Fundamentals of database systems , Pearson Education, 2006.
R2	Silberschatz, Korth and Sudarshan, Database system concepts , Tata McGraw Hill, 2004.
R3	ITL Education Solutions Ltd, Introduction to Database Systems , Pearson, 2010.
R4	Swapna Kishore & Rajesh Naik, SQL for professionals , Tata McGraw Hill, 2002.

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/106106093/
2	https://www.tutorialspoint.com/dbms/
3	https://www.studytonight.com/dbms/